University of Information Technology and Sciences

Department of Computer Science and Engineering
Course Title: Software Engineering and System Analysis Lab
Course Code: CSE 356
Section: 6B

Justification Form

Design and Implementation of a Real-World Software Engineering System (TouRz Travel Management Platform)

Student Information

Item	Information
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Course Title	Software Engineering and System Analysis Lab
Course Code	CSE 356
Section	6B1
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Instructor Information

Item	Information
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Designation	Lecturer
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Assignment Completion Justification Table

Ques No.	Task Description	Knowledge Profile (KP)	CEP Mapping	Method/Approach Used	Engineering Justification	Tools/Resources Used
Q1	Requirement analysis and stakeholder identification	K2, K3	P3, P6	Conducted stakeholder analysis (tourists, admins, system support). Captured functional/non-functional requirements.	Essential to establish clear system scope and prevent feature creep. Ensures all stakeholder needs (e.g., booking, chatbots) are aligned with system capabilities.	MS Word (for SRS), Google Forms, brainstorming sessions.
Q2	System modeling and technology selection	K2, K4	P3, P4	Evaluated SDLC models (Agile) and technology stacks. Selected PHP/MySQL over alternatives for rapid backend integration and deployment.	PHP offers direct compatibility with existing scripts (like chatbot_api.php and cancel_booking.php) and shared hosting, minimizing costs.	Web Servers (XAMPP), Git, VS Code.
Q3	System architecture and database design	K3, K5	P3, P7	Designed a 3-tier architecture. Created comprehensive UML models (Use Case, ERD, DFD, State, Sequence).	A 3-tier design separates the presentation, logic, and data layers, making the system modular, maintainable, and scalable.	PlantUML, Kroki API, Draw.io.
Q4	System implementation and module development	K4, K5, K6	P4, P7	Developed specific modules iteratively: Auth, Booking, Cancellations, and Chatbot support.	Iterative implementation allowed for continuous testing of complex intertwined modules without breaking the entire platform.	PHP, MySQL, HTML/CSS/JS, HTTP APIs.
Q5	Testing, validation, and debugging	K3, K4	P6, P7	Conducted unit testing on independent scripts (e.g., test cases for booking cancellations).	Prevented critical booking errors and ensured real-time database constraints were held (e.g., no double booking).	Postman, Browser DevTools, PHP Error Logging.

Ques No.	Task Description	Knowledge Profile (KP)	CEP Mapping	Method/Approach Used	Engineering Justification	Tools/Resources Used
				Performed integration testing.		
Q6	Project management and documentation	K5, K6, K7	P2, P6	Tracked milestones, prioritized modules based on dependencies, and documented the process (e.g., presentation slides).	Proper documentation and task breakdown ensured the project was completed on time, meeting the course outcomes safely and ethically.	Trello, Canva, Markdown, GitHub.

Engineering Design Decision Summary

Design Parameter	Selected Option	Reason for Selection	Performance Consideration	Reliability Consideration	Cost Consideration
System Architecture	3-tier Architecture	Clear separation of UI, business logic, and database.	Scales efficiently as user load increases.	High fault tolerance through modularity.	Low initial setup complexity.
Backend Technology	PHP	Native support for Apache/MySQL, great for web processing.	Fast response time for mid-scale platforms.	Highly proven and stable ecosystem.	Open-source and free.
Database System	MySQL	Strong relational integrity for booking management.	Fast read/write times for structured queries.	ACID compliance ensures no data loss.	Open-source and free.
Frontend Technology	HTML, vanilla CSS, JS	Offers max flexibility for responsive and dynamic design.	Lightweight, loads quickly without heavy frameworks.	Cross-browser compatibility.	Free tools and libraries.
Authentication Method	Session-based with Password Hashing	Secure standard for web applications to protect user privacy.	Low server overhead per request.	Protects against basic intrusion/hijacking.	Included natively in PHP.
Deployment Method	Localhost (XAMPP) /	Easiest path from	Adequate for testing and	Stable uptime for standard	Extremely cost-effective.

Design Parameter	Selected Option	Reason for Selection	Performance Consideration	Reliability Consideration	Cost Consideration
	Shared Web Hosting	development to production.	launch phases.	web traffic.	

KP Reflection Table

Knowledge Profile	How It Was Applied in the Assignment
K2 (Math, computing fundamentals)	Applied computational logic to handle dates, calculate booking costs, and manage inventory counting logic.
K3 (Engineering fundamentals)	Utilized Software Development Life Cycle (SDLC) principles for structured planning and requirement engineering.
K4 (Specialized engineering knowledge)	Implemented domain-specific code using PHP, interacting with APIs (like the chatbot), and designing complex SQL joins.
K5 (Engineering design knowledge)	Designed relational database schemas (ERDs) and UI/UX flows that prioritize usability for tourists.
K6 (Engineering practice & tech)	Leveraged modern tools: VS Code, Git for version control, and PlantUML for systematic UML generation.
K7 (Ethics, society, safety)	Implemented data privacy measures (secure hashing) and structured the system to handle data ethically and safely.

CEP Reflection Table

Complex Engineering Problem	How the Problem Was Addressed
P1	N/A (Not explicitly mapped for this project)
P2 (Conflicting technical issues)	Balanced the need for a rich, interactive UI against overall platform performance and loading speeds by optimizing frontend components.
P3 (No direct/obvious solution)	Engineered custom logic for the booking cancellation and refund rules (cancel_booking.php), which had no out-of-the-box solution.
P4 (Involves unfamiliar issues)	Integrated an AI/Chatbot API (chatbot_api.php) for user support, requiring independent research into API handling mechanisms.
P6 (Multiple stakeholders involved)	Designed distinct access layers and dashboards for tourists (booking), admins (management), and support systems.
P7 (Multiple interconnected sub-problems)	Successfully linked distinct modules: user authentication impacts booking permissions, which impacts the cancellation logic and capacity limits.

Student Declaration

I hereby declare that this assignment submission is my own work. All references and external resources used in this assignment have been properly acknowledged.

Student Signature:  ____________________________

Date: 19th May 2026